

Install Apache Airflow using uv (Simple & Fast Setup Guide)

Apache Airflow requires a supported Python 3 environment.

From Airflow 3.1.0 onwards, the following Python versions are supported:

 Python **3.10, 3.11, 3.12, 3.13**

Important Notes Before You Begin

- The officially supported installation method is using **pip** with constraints.
- Example (classic way using pip):

```
pip install "apache-airflow[celery]==3.0.0" \
--constraint "https://raw.githubusercontent.com/apache/airflow/constraints-3.0.0/constraints-3.10.txt"
```

- Tools like **Poetry** or **pip-tools** are *not officially supported* because they handle dependencies differently from **pip/uv**, especially regarding **constraints vs. requirements**.
 - If you use those tools, you must manually adapt Airflow's constraint files to their format.
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Now: Installing Airflow with **uv** (Recommended Modern Way)

uv is a fast Python package manager and environment tool. It makes installing Airflow smooth, reproducible, and faster than traditional **pip**.

Step 1: Install uv (if you haven't already)

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Or using **pip**:

```
pip install uv
```

✅ Step 2: Create and activate a Python 3.10+ environment

```
uv venv --python=3.10
source .venv/bin/activate # Linux/Mac
# or
.venv\Scripts\activate   # Windows PowerShell
```

✅ Step 3: Install Apache Airflow (using constraints via **uv**)

```
uv pip install "apache-airflow[celery]==3.1.0" \
--constraint "https://raw.githubusercontent.com/apache/airflow/constraints-3.1.0/constraints-3.10.txt"
```

✅ Step 4: Verify Installation

```
airflow version
```

The expected output will show the installed Airflow version.